Milk powders — whole, skim, and specialty: HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified standard for milk powders — whole, skim, and specialty — stated as sold on a powder basis. Powder carries its own government maximum levels reflecting the concentrated matrix; there is no reconstitution conversion. Every limit is the strictest such sovereign maximum, with a disclosed read-across where none regulates the exact analyte.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; no dairy reconstitution; disclosed read-across; species/packaging controls)

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS
Lead (Pb)	20	human-grade read-across (Israel)
Cadmium (Cd)	1,000	government maximum level
Total arsenic (tAs)	500	government maximum level

Inorganic arsenic (iAs)	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	50	government maximum level
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	250	human-grade read-across (European Union)
Tin (Sn)	250,000	human-grade read-across (Singapore)
Aluminium (Al)	500	human-grade read-across (Australia / New Zealand)
Total chromium (Cr)	2,000	government maximum level
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	20	Israel — human-food analog
Cadmium (Cd)	1,000	government maximum level
Total arsenic (tAs)	500	government maximum level
Inorganic arsenic (iAs)	≤ tAs	iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell.
Mercury, total (tHg)	50	government maximum level
Methylmercury (MeHg)	≤ tHg	MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell.
Nickel (Ni)	250	European Union — human-food analog
Tin (Sn)	250,000	Singapore — human-food analog
Aluminium (Al)	500	Australia / New Zealand — human-food analog

Total chromium (Cr)	2,000	government maximum level
Hexavalent chromium (Cr-VI)	≤ Cr	Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell.

4 | Cite this standard

Pendergrass, K. (2026). *Milk powders — whole, skim, and specialty: HMTc standard* (Standard No. 10.83138/hmtc.std.24.dairy-milk-powders). Institute of Contaminant Standards. https://doi.org/10.83138/hmtc.std.24.dairy-milk-powders

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References

[1] Australia / New Zealand — human-food maximum levels
Binds Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] Israel — human-food maximum levels
Binds Pb. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[4] Singapore — human-food maximum levels
Binds Sn. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/dairy-milk-powders. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

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